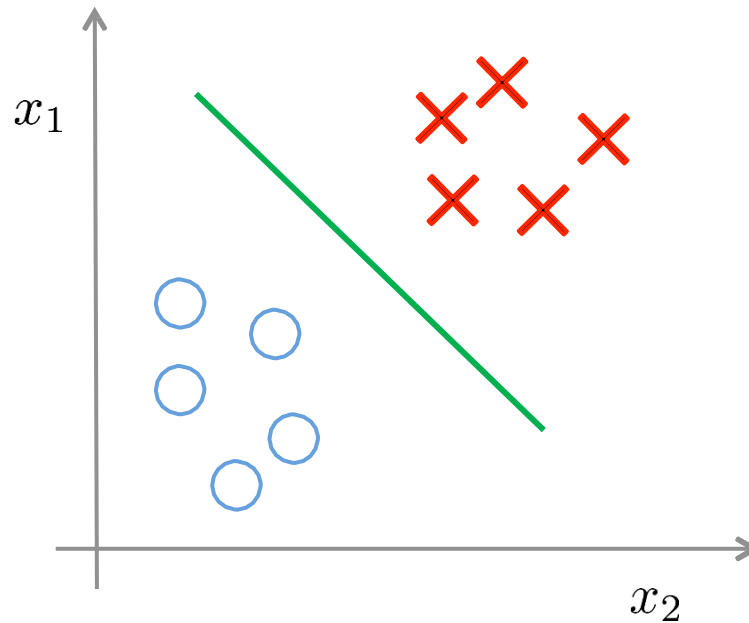


Clustering

Prof. Giuseppe Serra

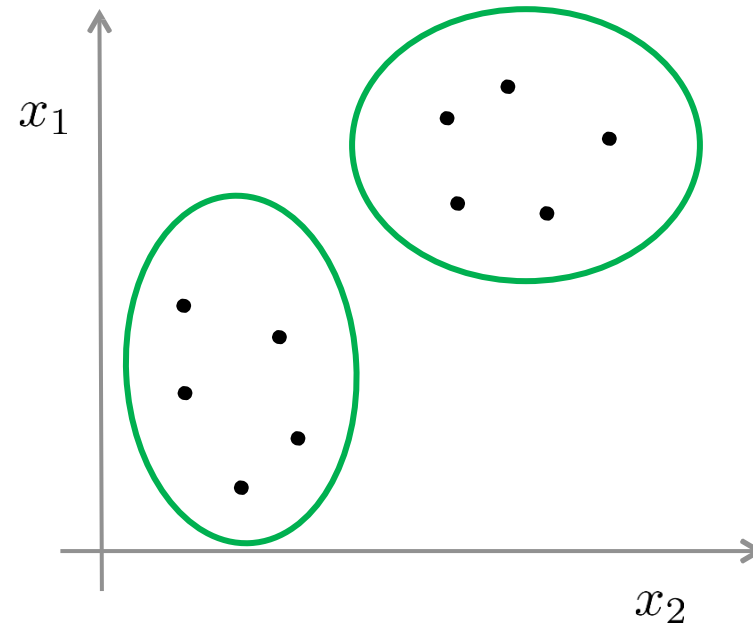
Università degli Studi di Udine

Supervised Learning



Training set: $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), (x^{(3)}, y^{(3)}), \dots, (x^{(m)}, y^{(m)})\}$

Unsupervised Learning

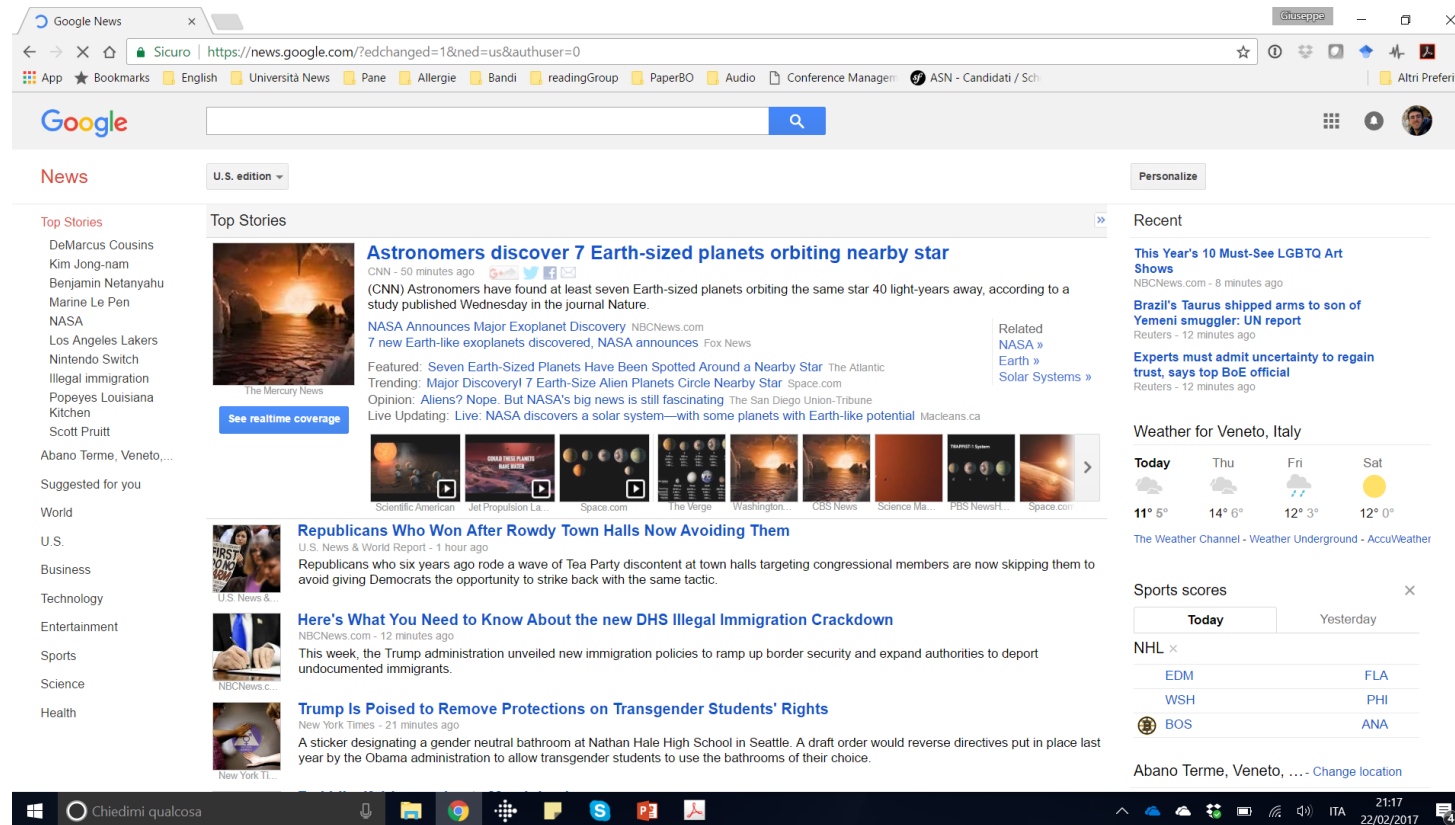


Training set: $\{x^{(1)}, x^{(2)}, x^{(3)}, \dots, x^{(m)}\}$

Google News

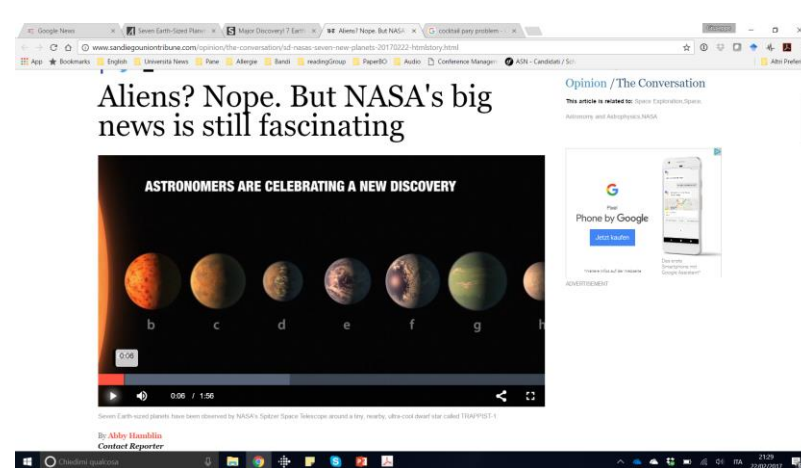
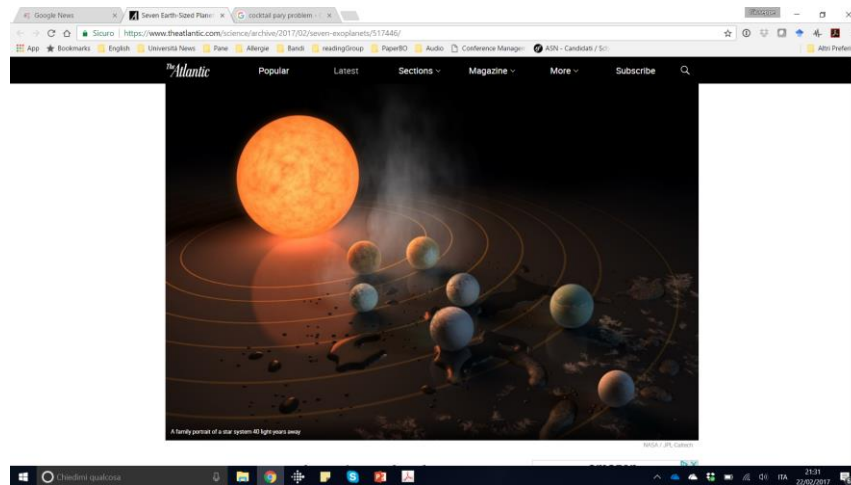
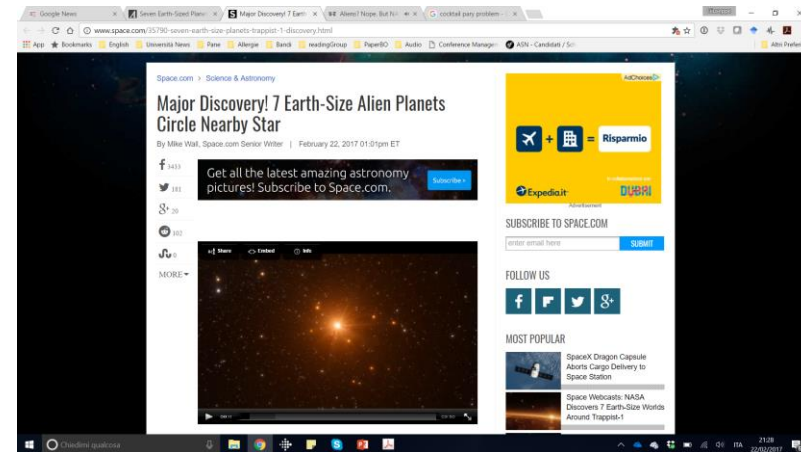
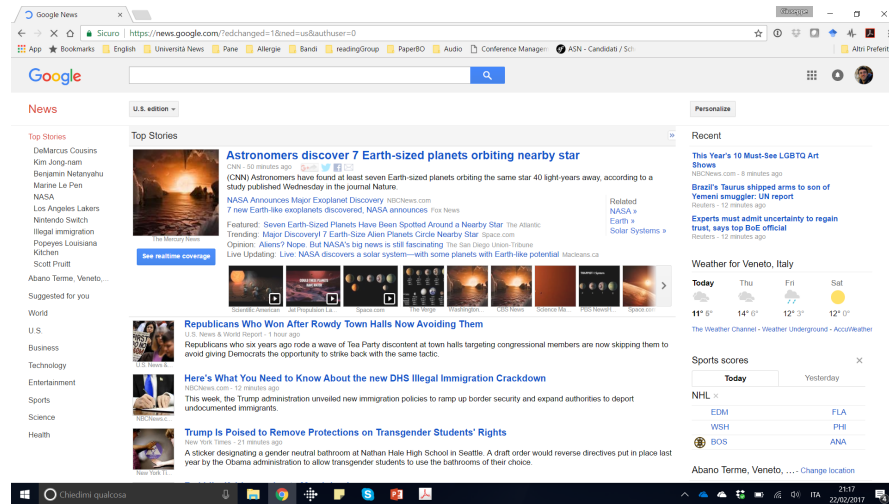
Cluster algorithms can be used in different applications

Everyday Google News analyses thousands of thousands of new stories and groups them into cohesive stories.

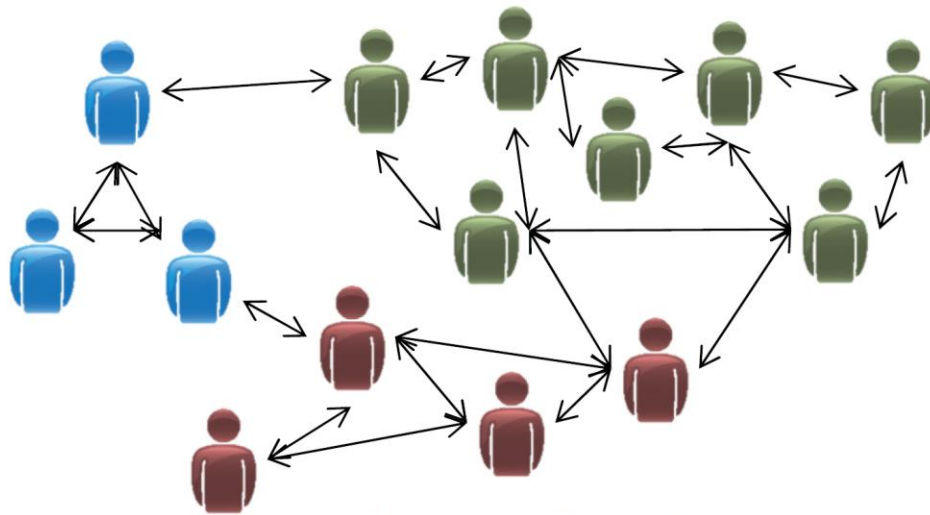


The screenshot shows the Google News homepage. The main content area features a large story titled "Astronomers discover 7 Earth-sized planets orbiting nearby star" with a video player and a "See realtime coverage" button. Below this, there are smaller stories including "Republicans Who Won After Rowdy Town Halls Now Avoiding Them" and "Here's What You Need to Know About the new DHS Illegal Immigration Crackdown". The left sidebar lists categories like "World", "U.S.", "Business", "Technology", "Entertainment", "Sports", "Science", and "Health". The right sidebar shows "Recent" news, "Weather for Veneto, Italy", and "Sports scores". The bottom of the page shows a Windows taskbar with various application icons and the system clock.

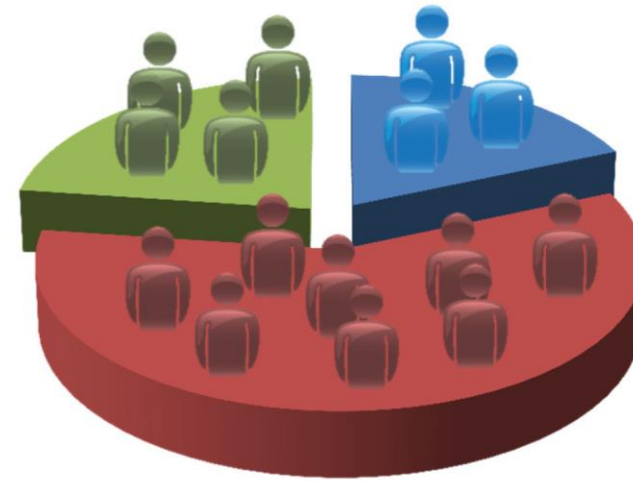
Google News



Other Examples



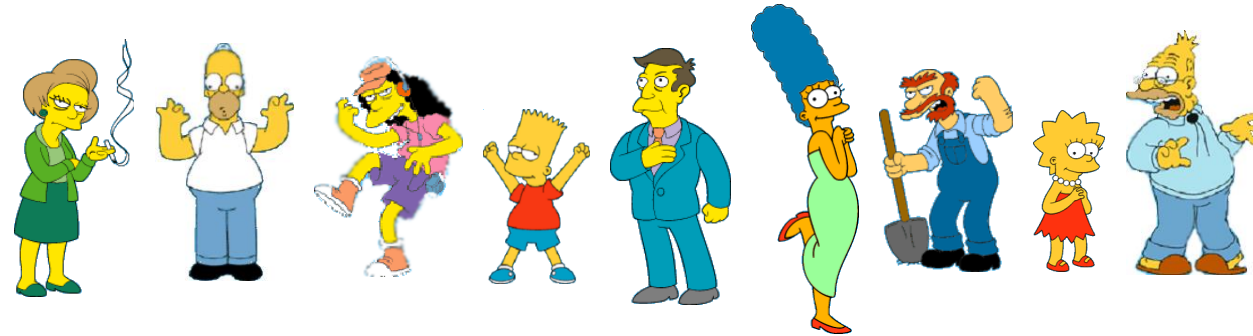
Social network analysis



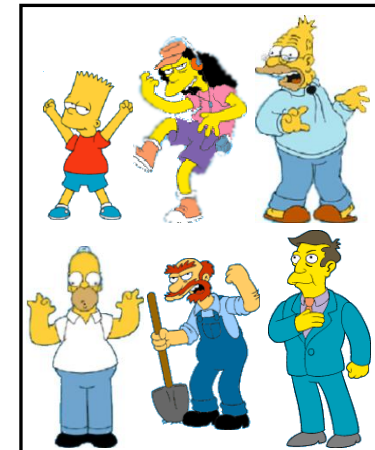
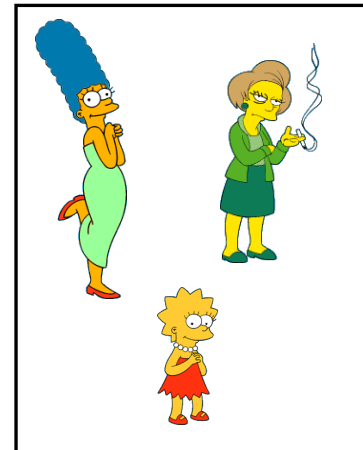
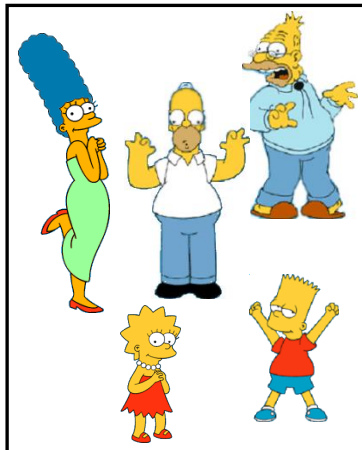
Market segmentation

Clustering

What is a natural grouping among these objects?

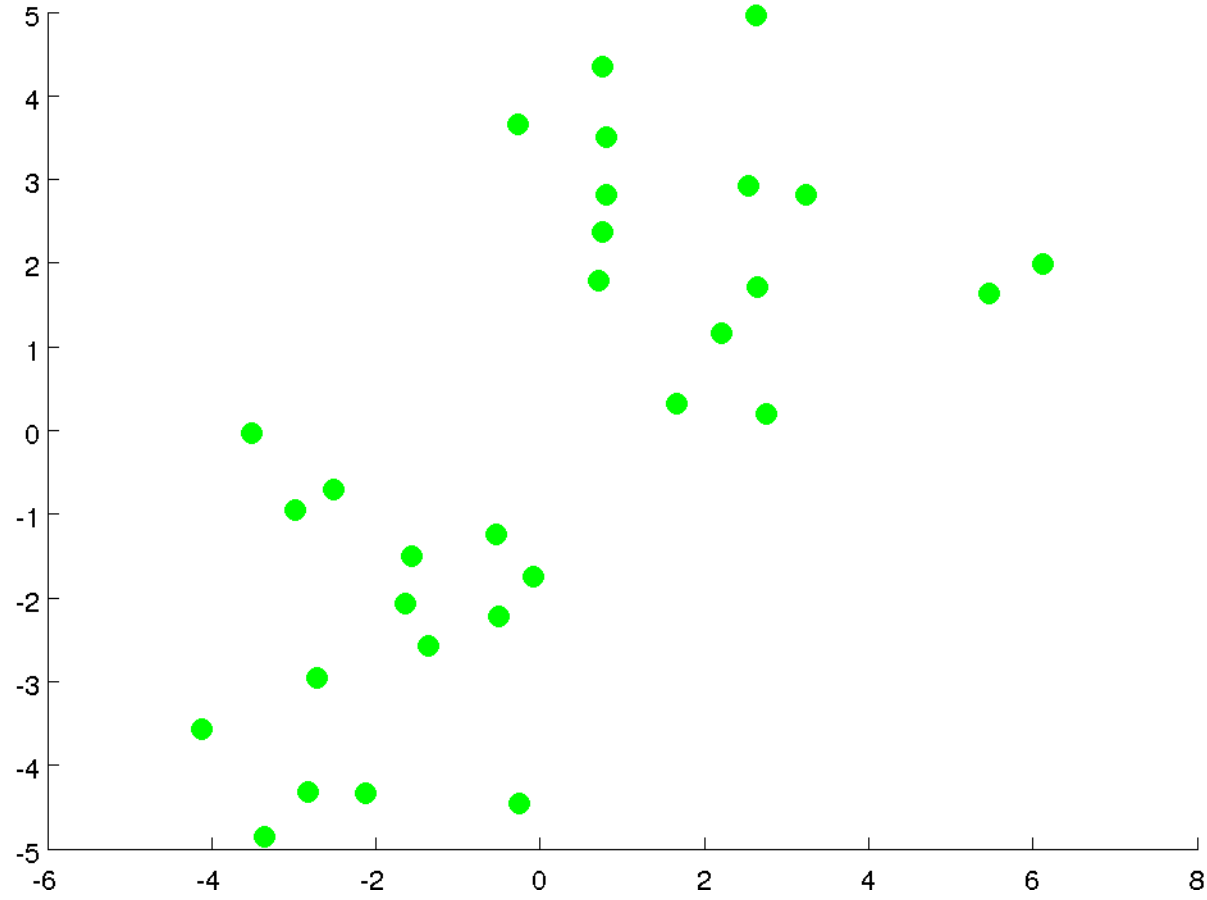


Clustering is subjective

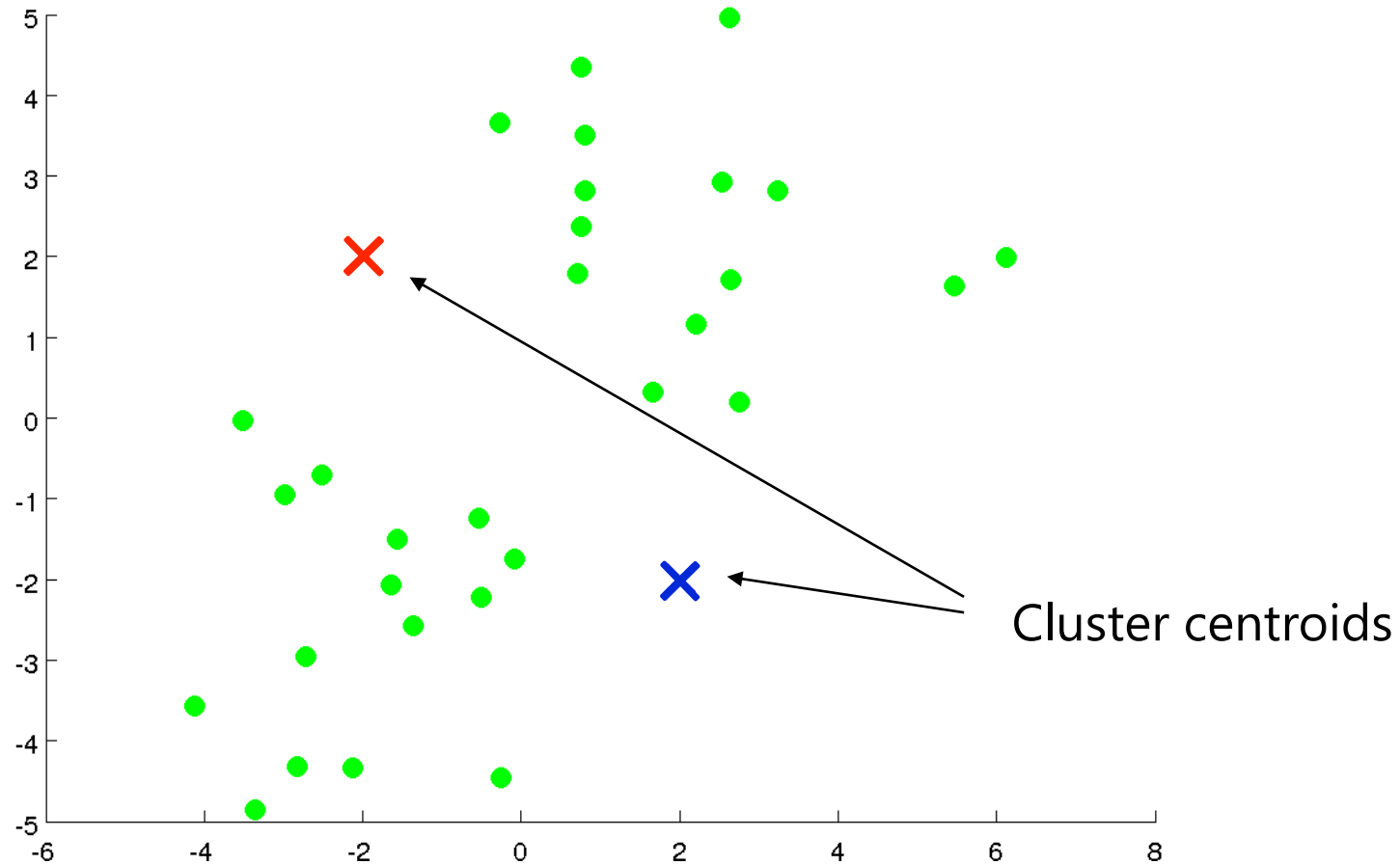


K-means

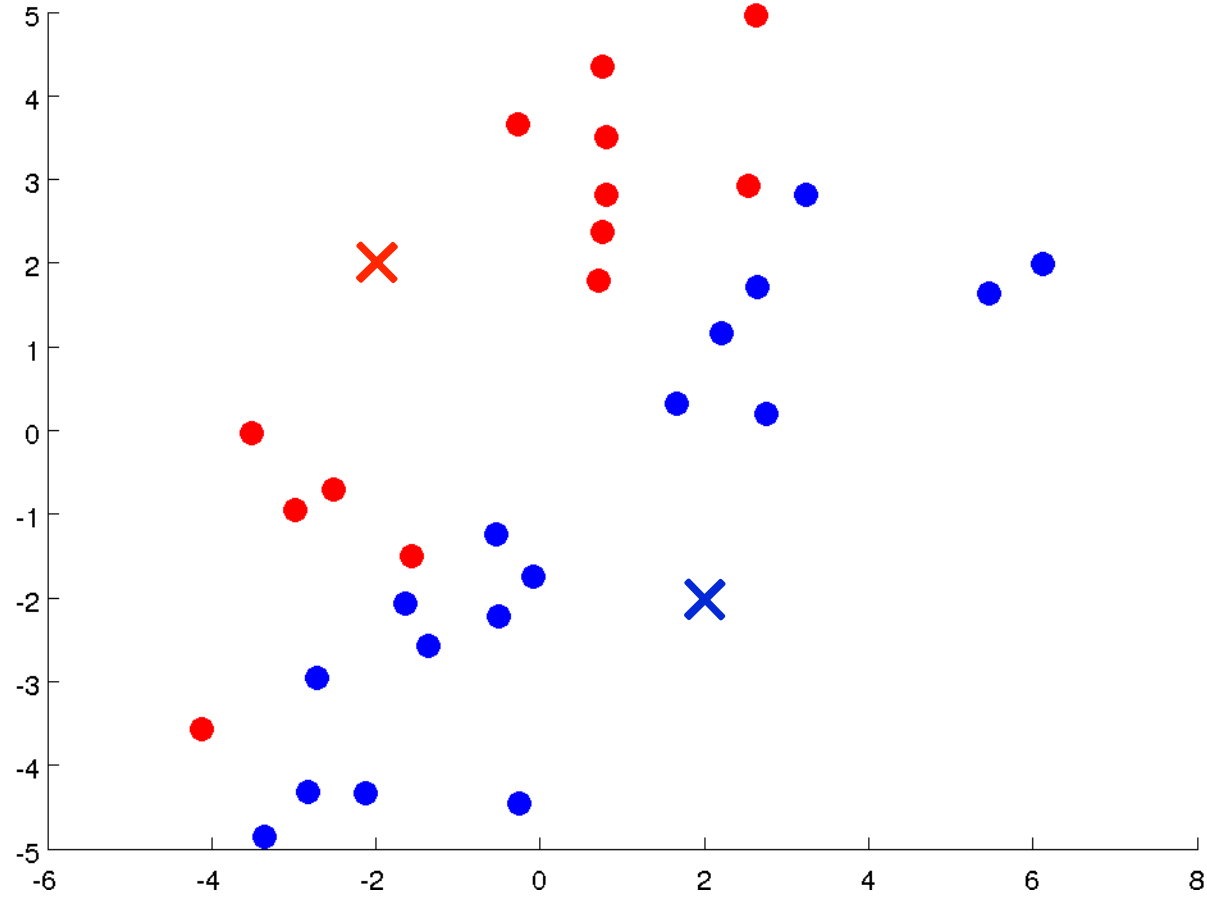
K-Means



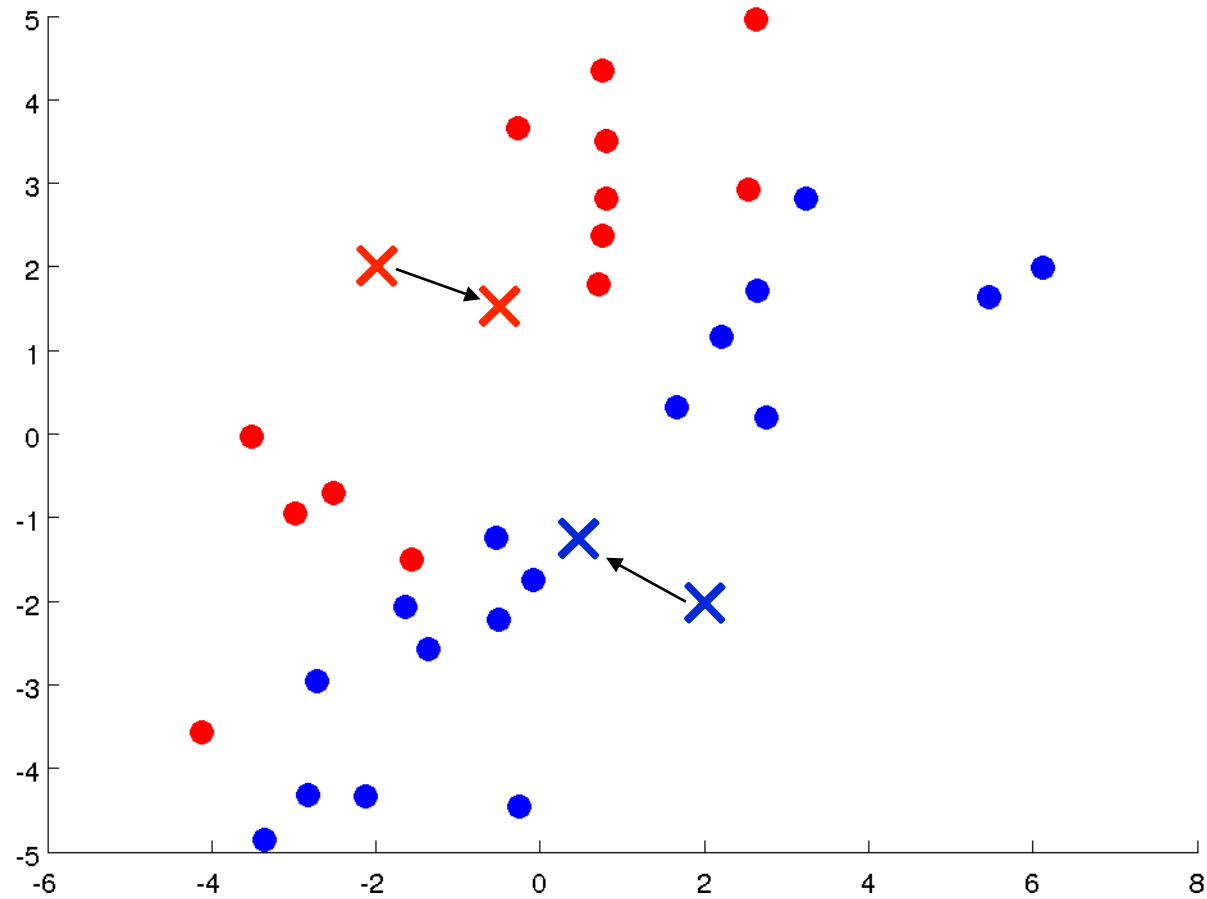
K-Means



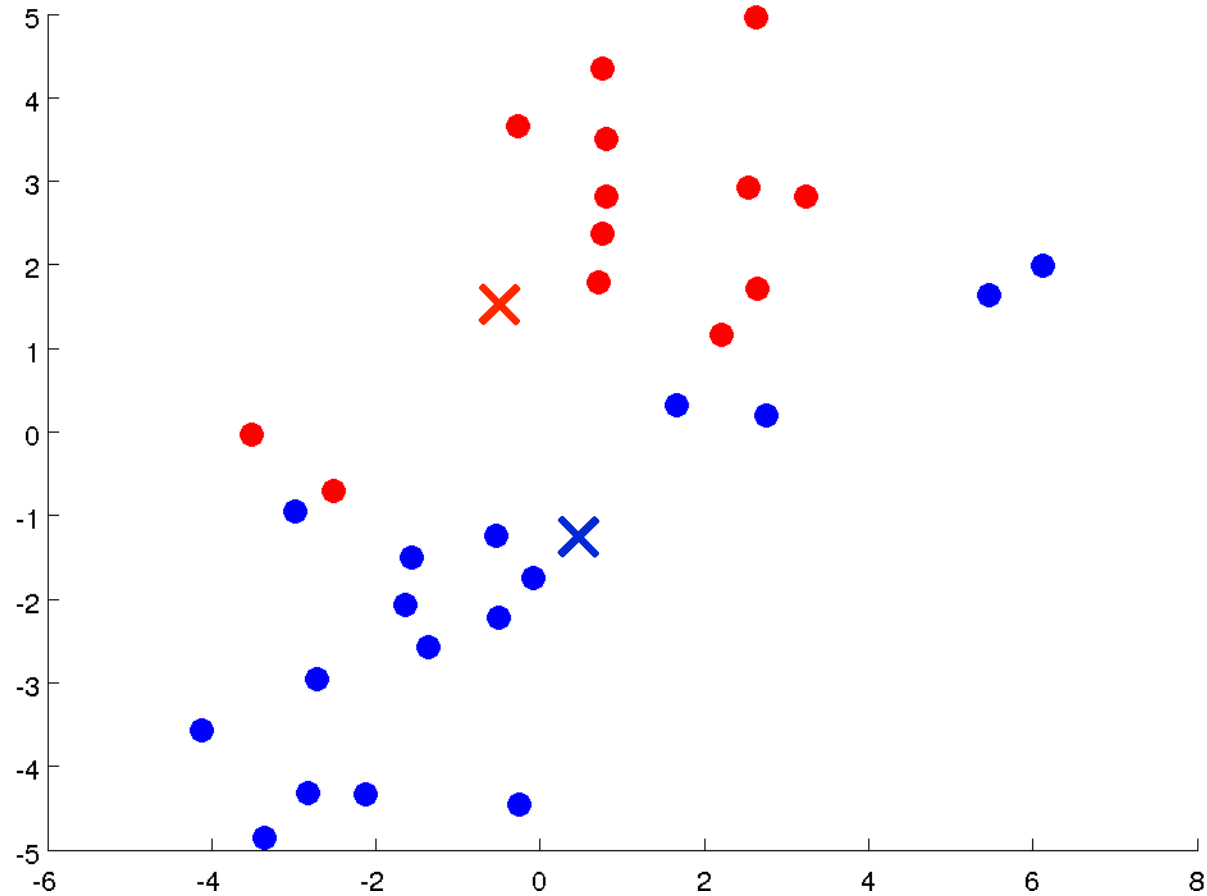
K-means



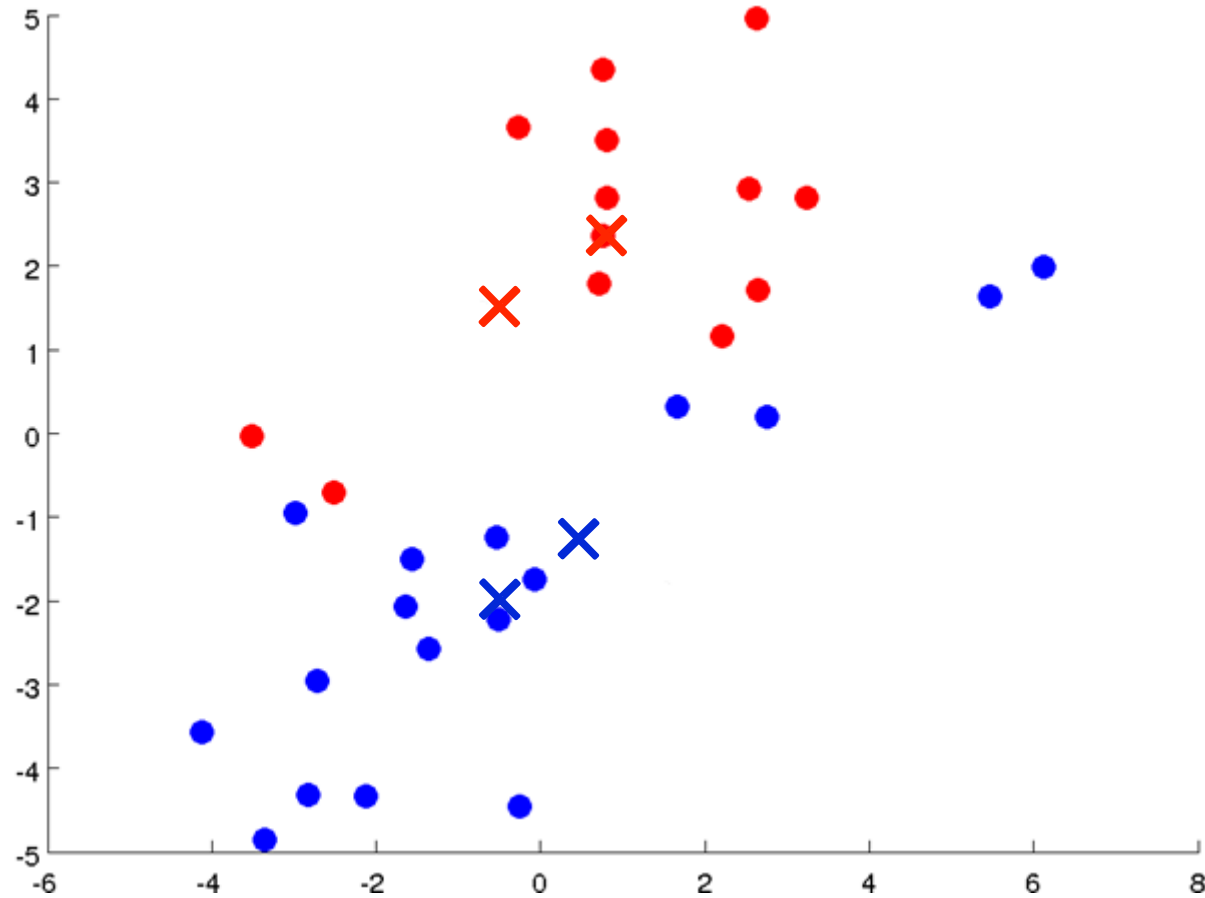
K-means



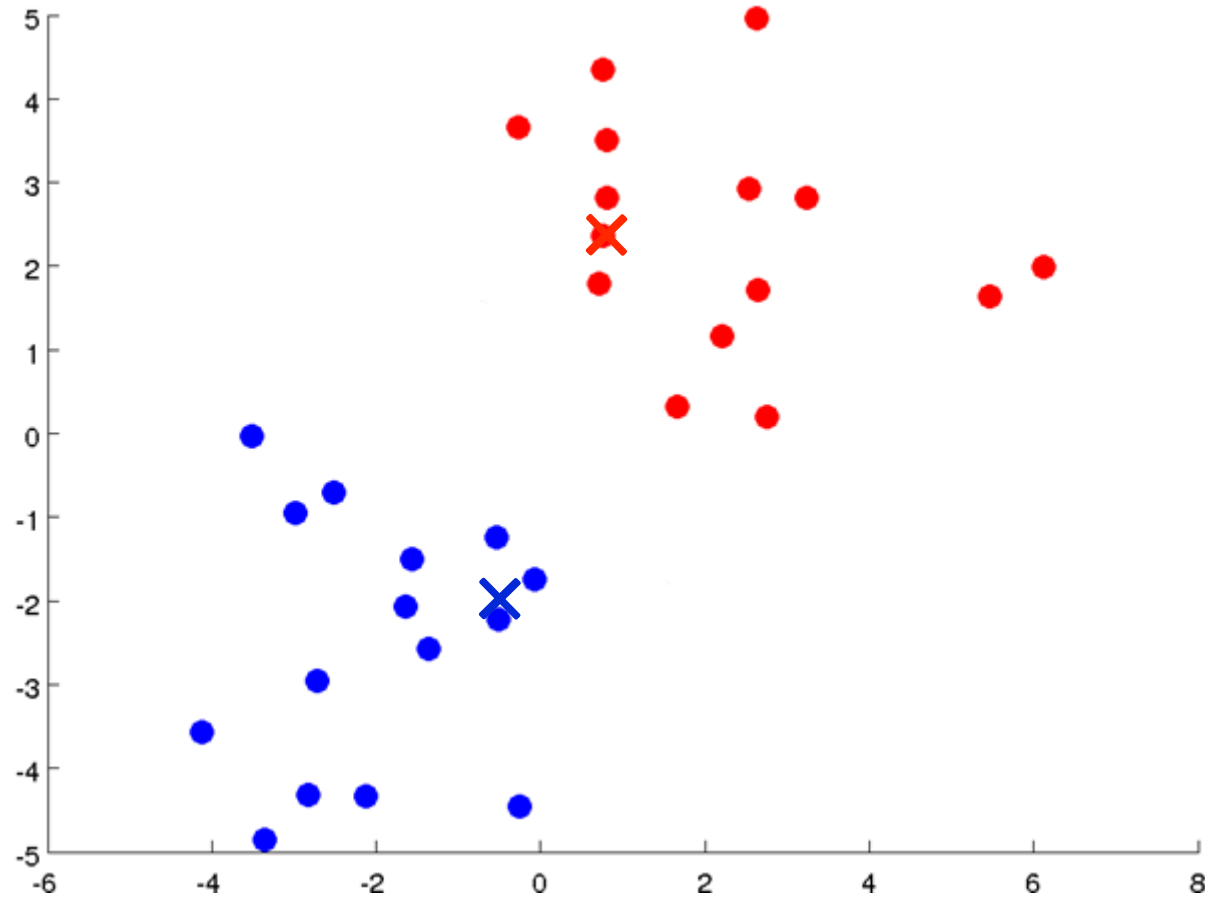
K-means



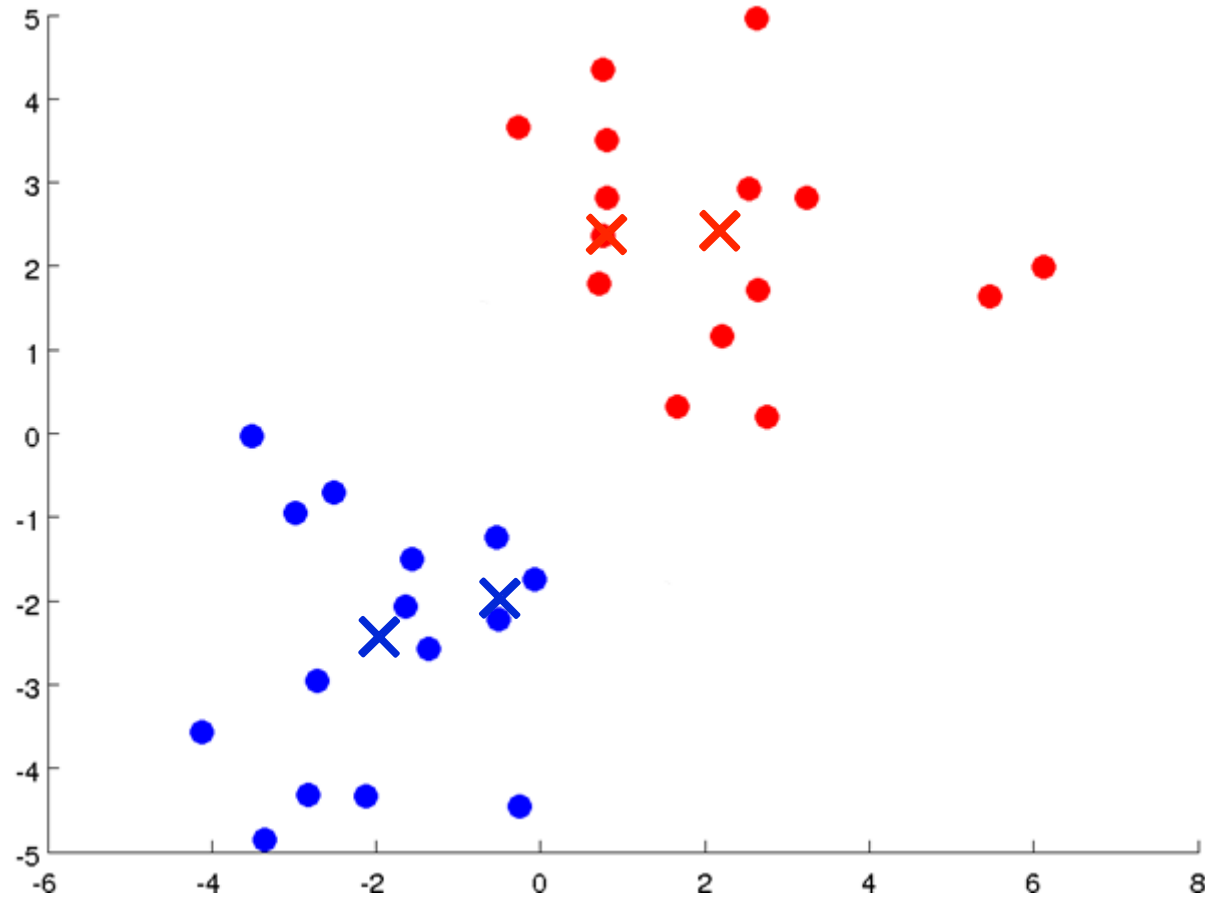
K-means



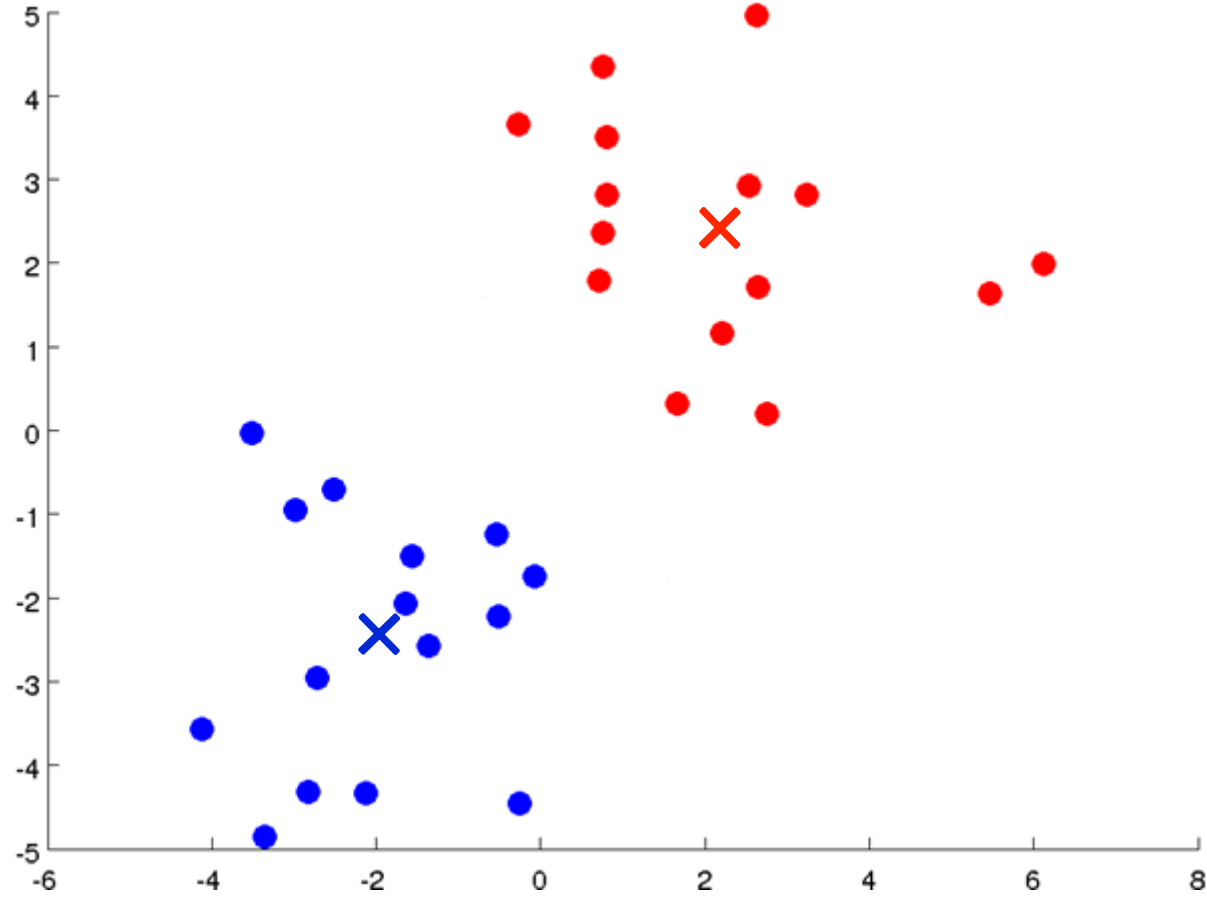
K-means



K-means



K-means



K-means Algorithm

Given K (number of cluster) and the Training set $\{x^{(1)}, x^{(2)}, \dots, x^{(m)}\}$

Randomly initialize K cluster centroids $\mu_1, \mu_2, \dots, \mu_K \in \mathbb{R}^n$

Repeat {

 for $i = 1$ to m

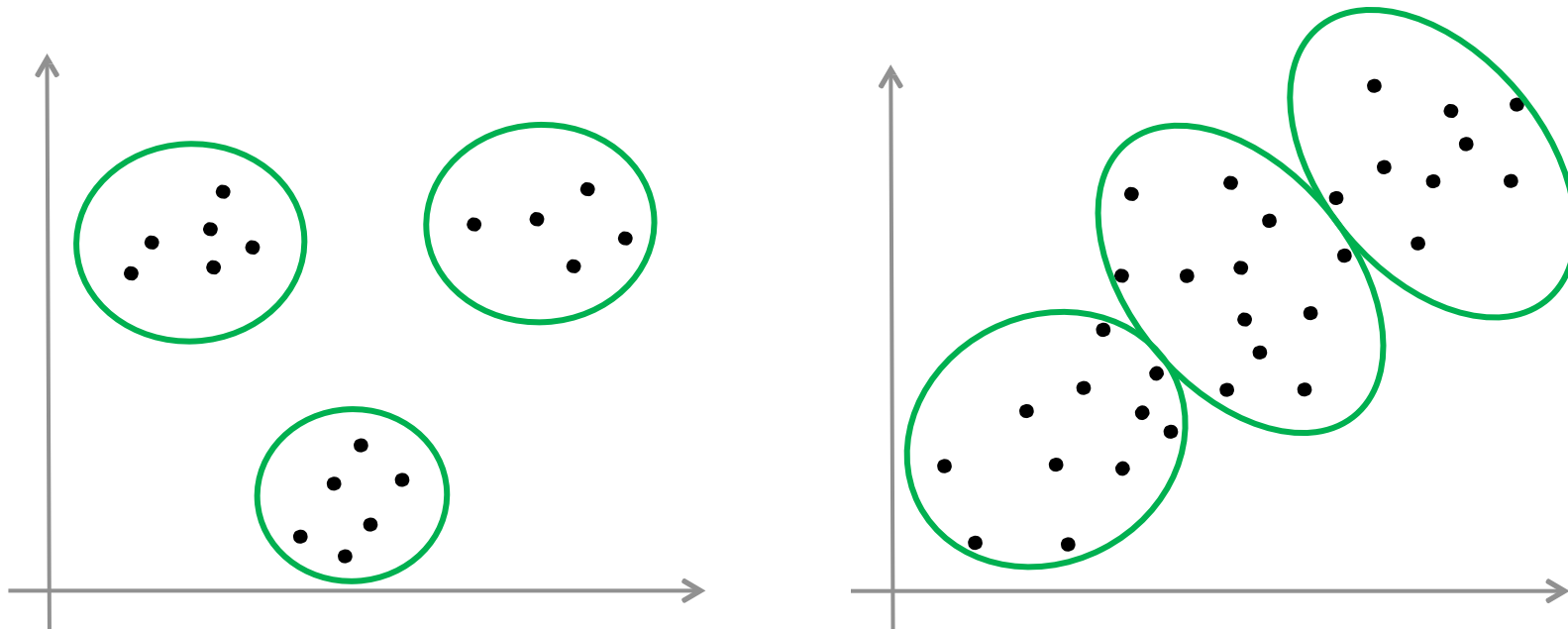
$c^{(i)} :=$ index (from 1 to K) of cluster centroid
 closest to $x^{(i)}$

 for $k = 1$ to K

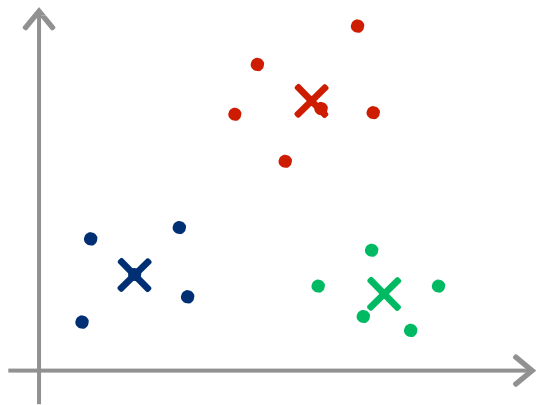
$\mu_k :=$ average (mean) of points assigned to cluster k

}

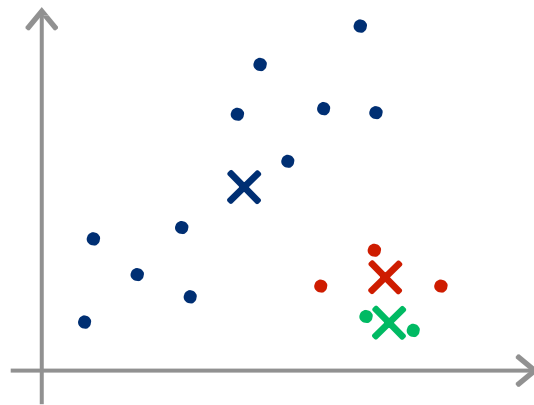
K-means for non-separated clusters



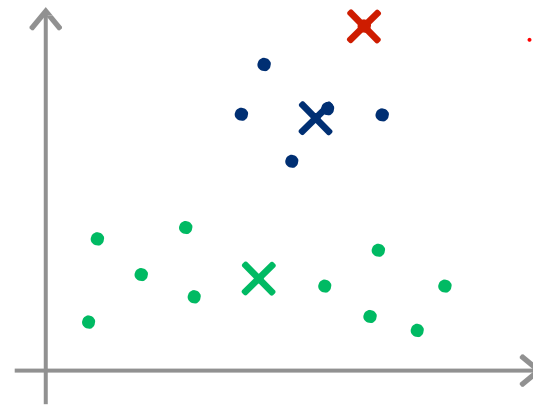
Local optimal



Case 1

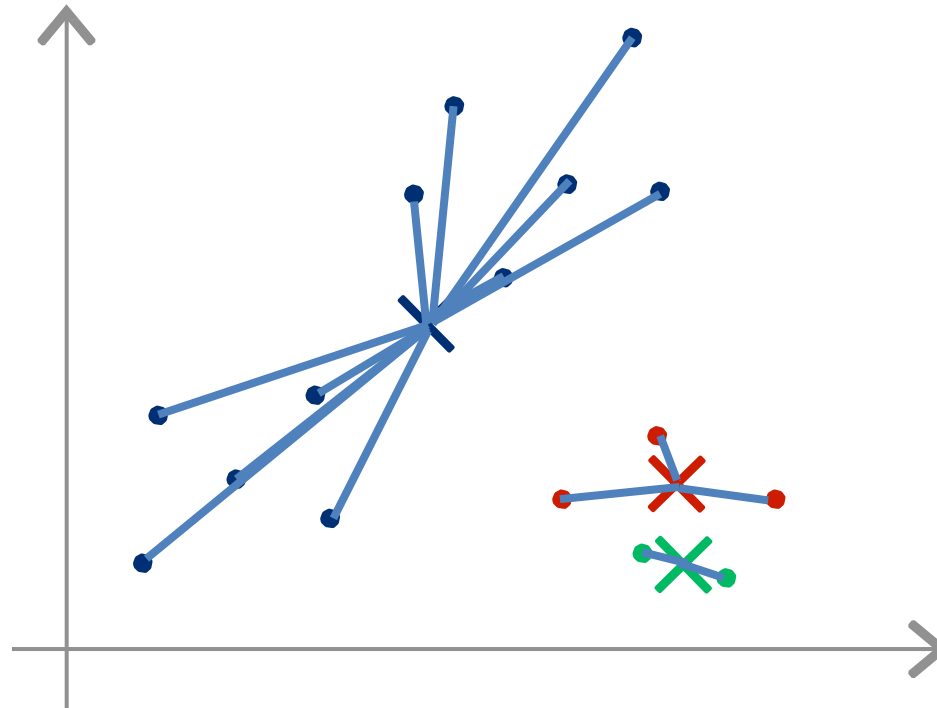


Case 2



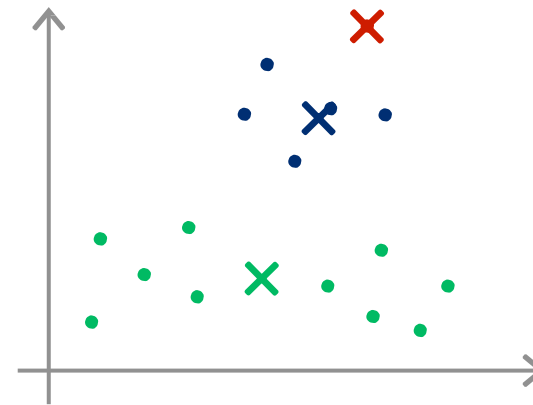
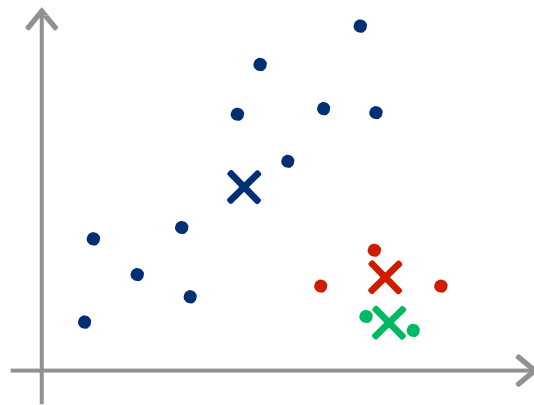
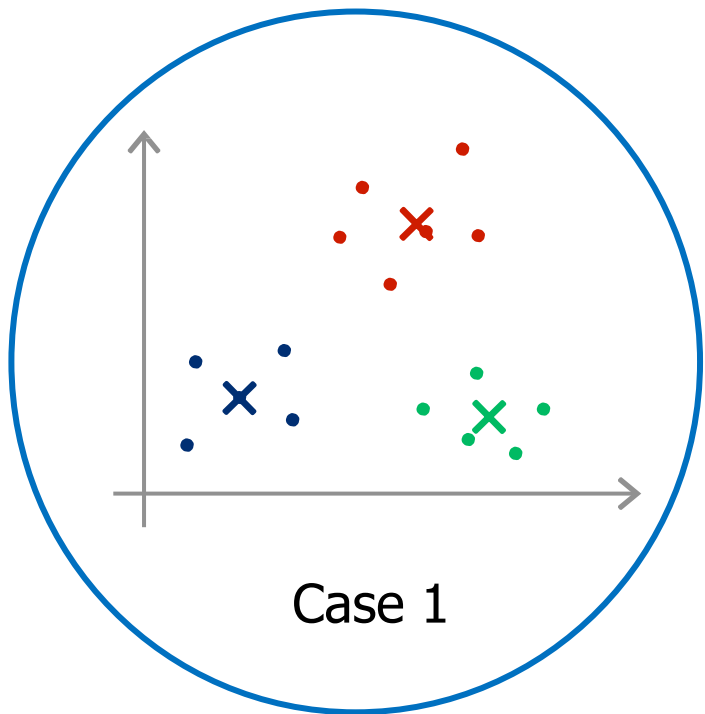
Case 3

Cost Function



The cost function is the sum of lengths of the blue segments

Local optimal



Random Initialization

For $i = 1$ to 100 {

 Randomly initialize K-means.

 Run K-means. Get $c^{(1)}, \dots, c^{(m)}, \mu_1, \dots, \mu_K$.

 Compute the cost function

}

Pick clustering that gave lowest cost

Hierarchical Agglomerative Clustering

Hierarchical Agglomerative Clustering

We start with every data point in a separate cluster

We keep merging the most similar pairs of data points/clusters until we have one big cluster left

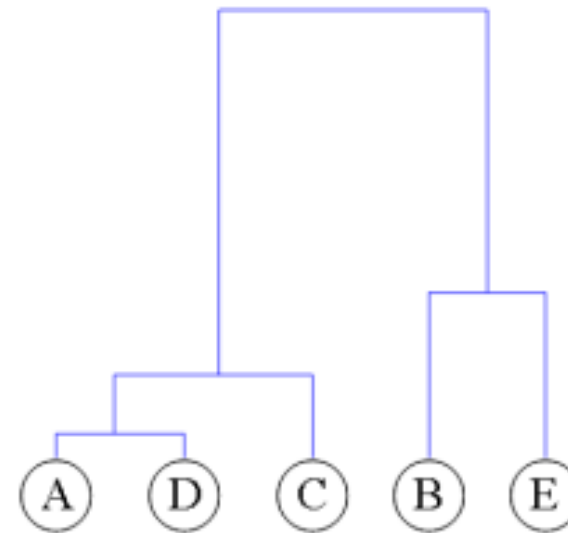
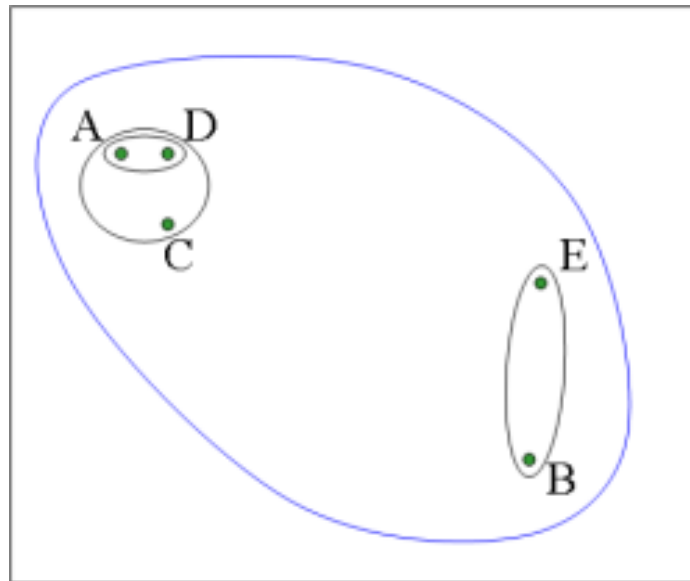
This is called a bottom-up or agglomerative method

Hierarchical Agglomerative Clustering

This produces a binary tree or dendrogram

The final cluster is the root and each data item is a leaf

The height of the bars indicate how close the items are

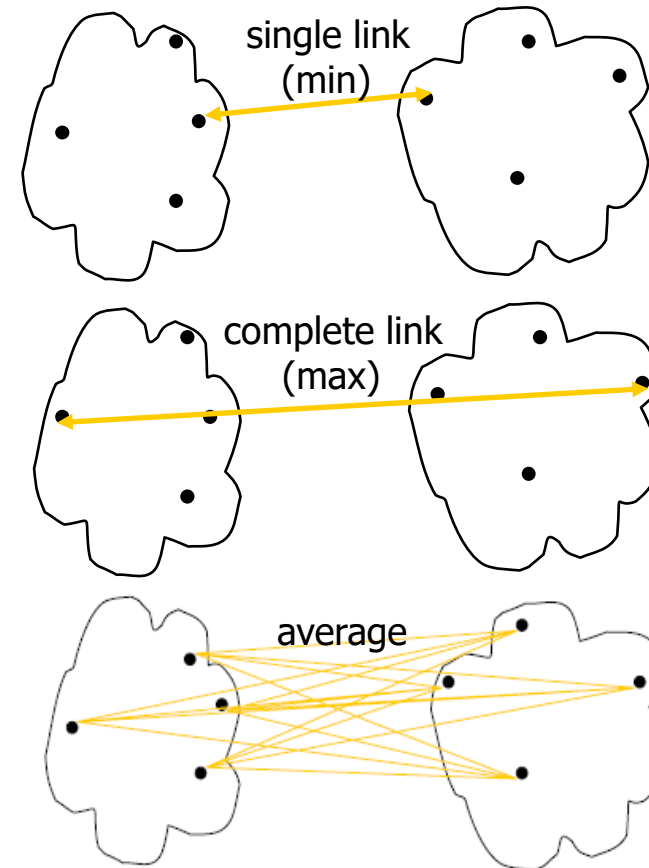


Cluster Distance Measures

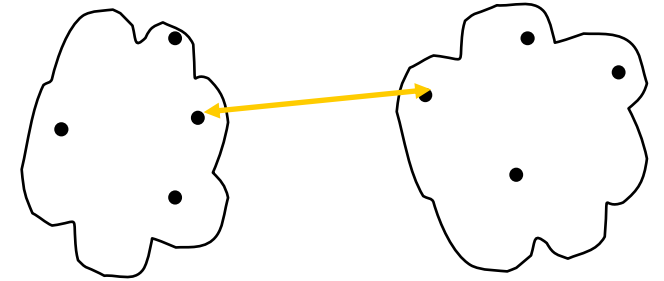
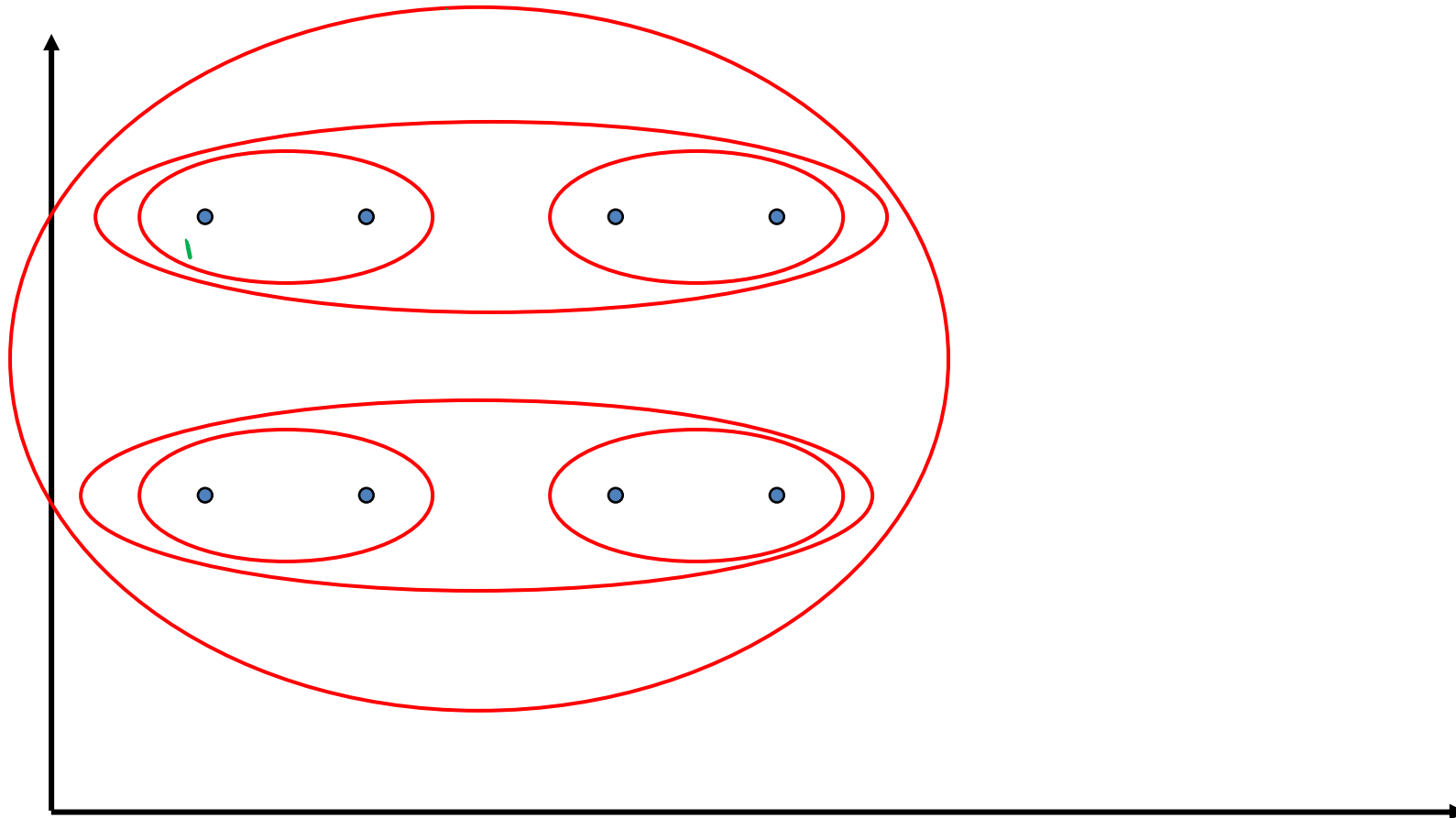
Single link: smallest distance between an element in one cluster and an element in the other

Complete link: largest distance between an element in one cluster and an element in the other

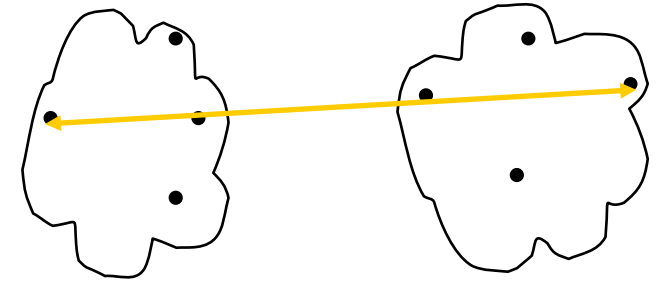
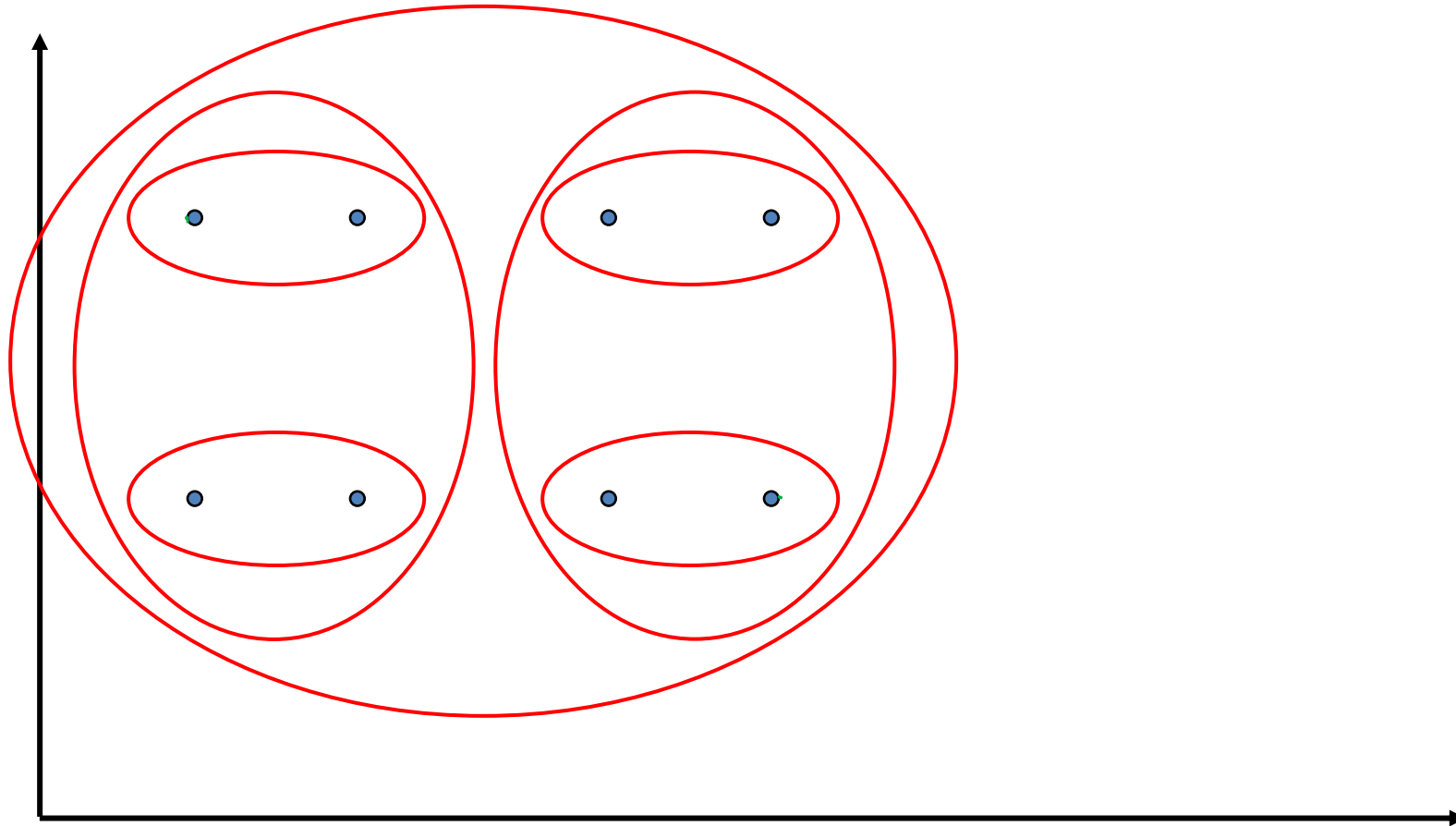
Average: avg distance between elements in one cluster and elements in the other



Single Link Example



Complete Link Example



DBSCAN Algorithm

Density-Based Clustering Methods

Clustering based on density (local cluster criterion), such as density-connected points or based on an explicitly constructed density function

Major features:

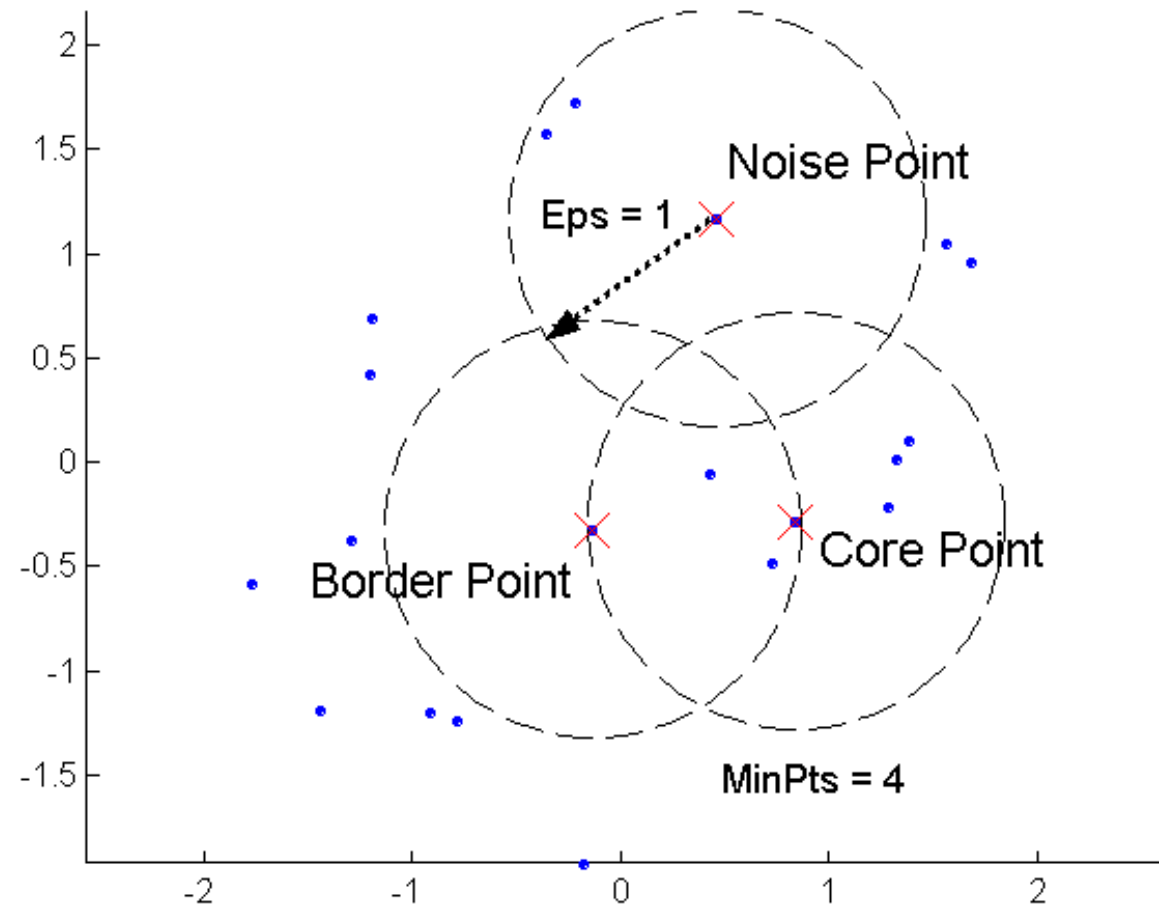
- Discover clusters of arbitrary shape
- Handle noise
- One scan
- Need density parameters

DBSCAN is a Density-Based Clustering

Nice visualization

<https://www.naftaliharris.com/blog/visualizing-dbscan-clustering/>

DBSCAN: Core, Border, and Noise Points.



DBSCAN is a density-based algorithm

Density = number of points within a specified radius r (Eps)

A point is a **core point** if it has more than a specified number of points (MinPts) within Eps

- These are points that are at the interior of a cluster

A **border point** has fewer than MinPts within Eps, but is in the neighborhood of a core point

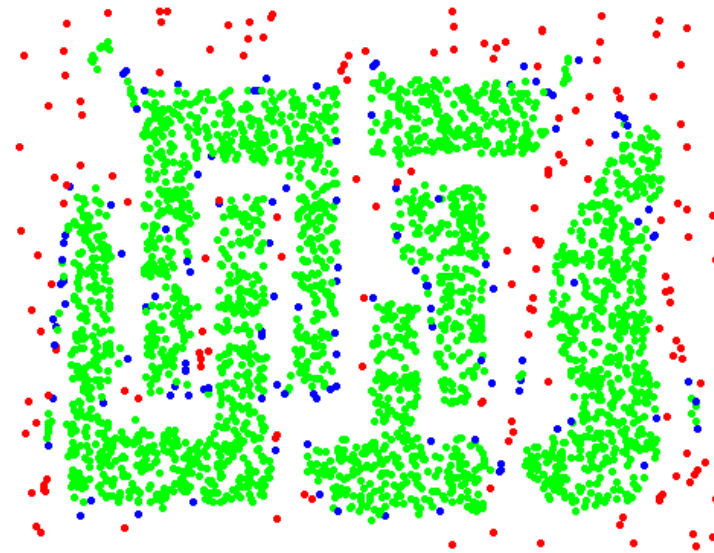
A **noise point** is any point that is not a core point or a border point. Noise points are discarded

Any two core points are close enough– within a distance Eps of one another – are put in the same cluster

DBSCAN: Core, Border, and Noise Points



Original Points



Point types: **core**,
border and **noise**

Eps = 10, MinPts = 4

DBSCAN pseudo Algorithm

Select a point p

Retrieve all points from p wrt Eps and $MinPts$.

If p is a noise point is labelled as noise

If p is a border/core point, a cluster is formed.

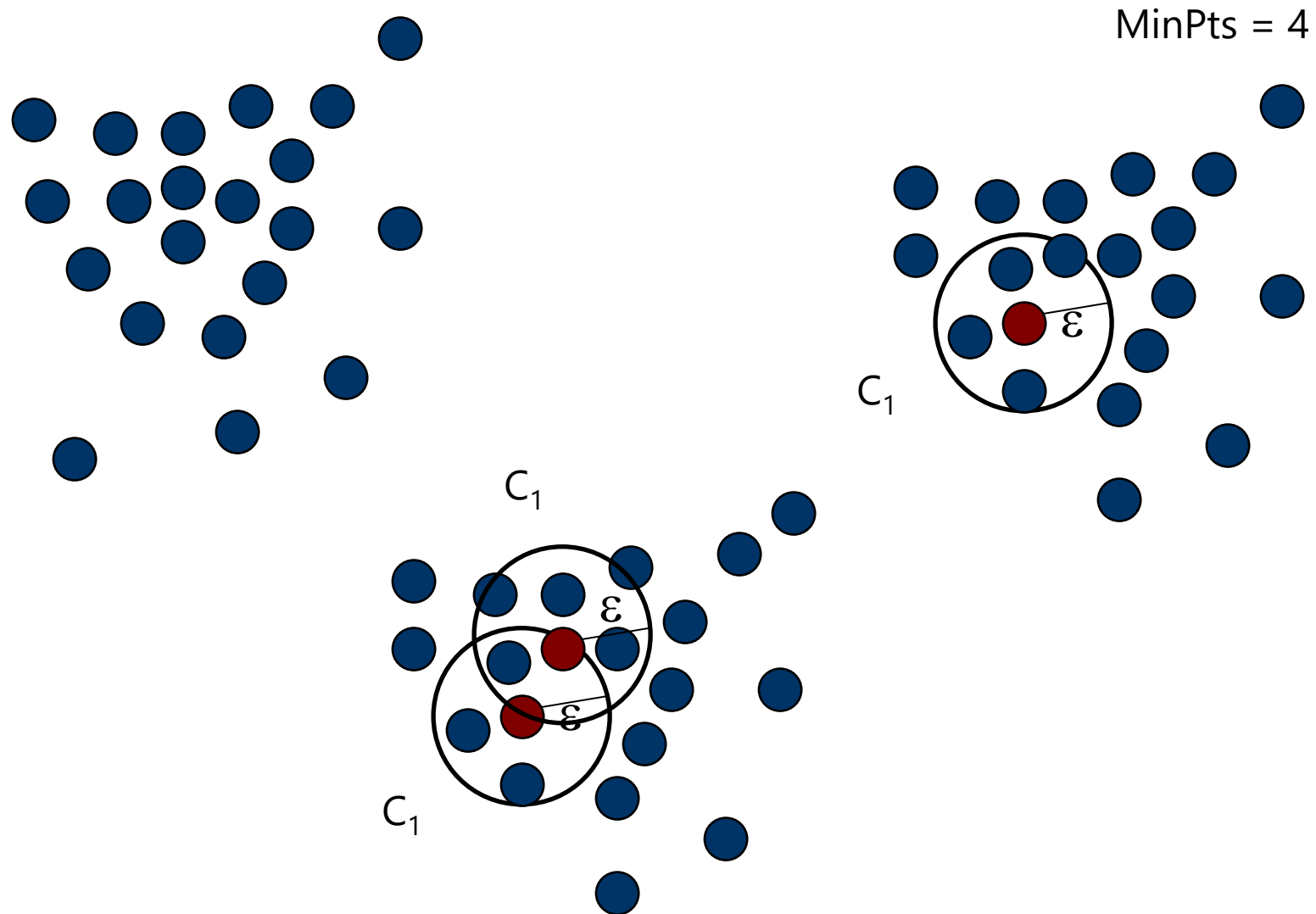
All the neighbors (with Eps) of the core point are visited.

- If they have not be assigned a cluster yet, they are assigned the new cluster label that was just created.
- If they are core sample, their neighbors are visited in turn

Continue the process until there are no more core samples with distance Eps of the cluster.

Another point that has not yet been visited is picked and the same procedure is repeated.

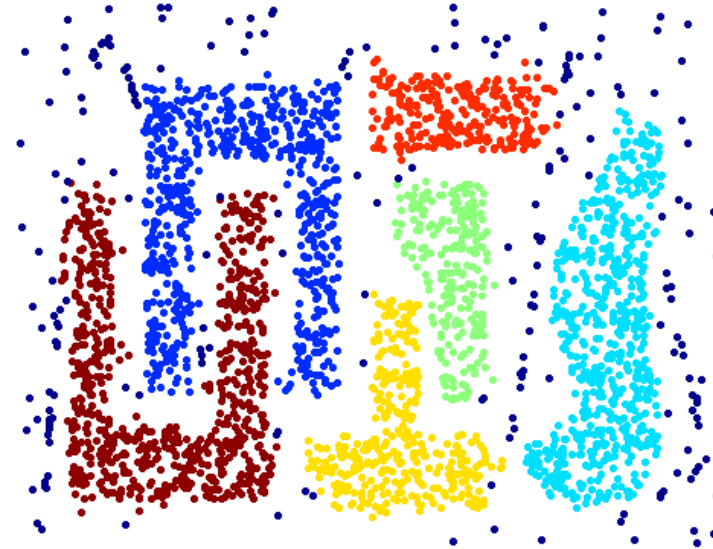
Example



When DBSCAN works well



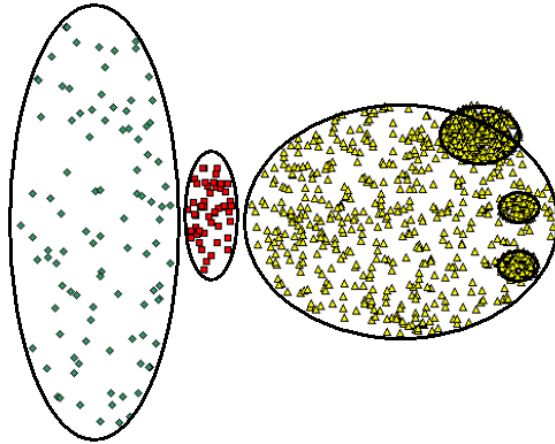
Original Points



Clusters

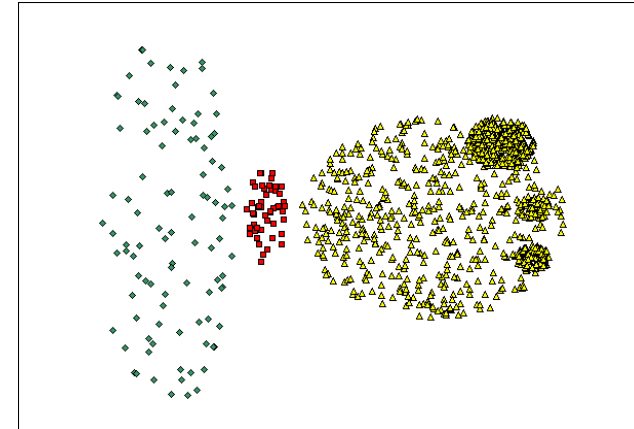
- Resistant to Noise
- Can handle clusters of different shapes and sizes

When DBSCAN does not work well

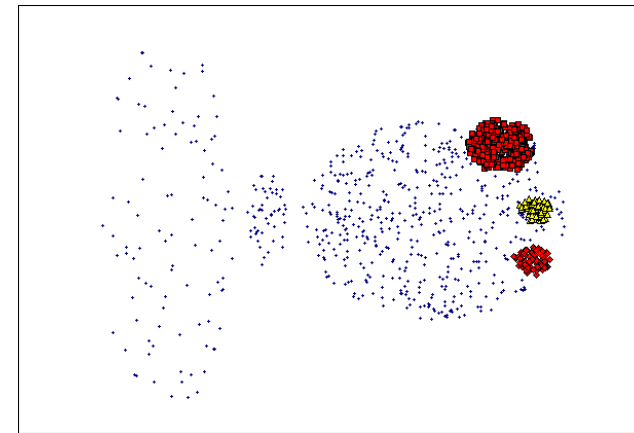


Original Points

- Varying densities



(MinPts=4, Eps=9.75).



(MinPts=4, Eps=9.92)